

Observation report about difference between spectra of original and rotated image

Comparing rotated image spectra with original image's spectra, and writing observations

- 1.) after rotating the image 90 degrees anti-clockwise, the 2D Fourier transform magnitude spectrum also rotates by 90 degrees in the same direction.
- 2.) The intensity distribution remains unchanged, but the orientation of the frequency components shifts, confirming that the Fourier transform
- 3.) preserves the structure of an image in the frequency domain while aligning it to the image's spatial orientation. This property illustrates the

close link between geometric transformations in space and their mirror in the frequency world.

Theory explaining it in detail

We take a grayscale image—let's say a cat's face—and compute its 2D Fourier Transform to observe how the frequency components are distributed across the spatial frequencies. The result is a magnitude spectrum that highlights where the image carries significant structural or edge information. Typically, this spectrum has the low-frequency components at the centre (after applying `fftshift`) and the high-frequency textures radiating outward.

Now, when we rotate the input image anti-clockwise by 90 degrees and compute its 2D Fourier Transform, something fascinating happens.

The **entire magnitude spectrum also rotates** by 90 degrees in the **same direction** as the image. This reveals a fundamental property of the Fourier transform:

Rotation in the spatial domain causes an equivalent rotation in the frequency domain.

However, while the orientation of the spectrum changes, the **content itself does not**. The intensity, symmetry, and energy distribution remain the same—just rotated. For example, if the original image had more horizontal features (like whiskers or stripes), those frequencies appear along the horizontal axis in the original spectrum. After rotating the image, those features now run vertically, and so the energy in the spectrum shifts accordingly, aligning along the vertical axis.

This means that **the structure of the spectrum is preserved**, but its **directional alignment** changes to reflect the rotation of the image. The center of the spectrum, which holds the low-frequency content (general shape, brightness), remains at the center, but the patterns around it rotate.